2015 (JUNE)
MCA SECOND SEMESTER
MCA-204
COMPUTER ORGANISATION-II

Full Marks: 75.
Time: 3 Hours.

The figures in the margin indicate full marks for the questions. ***Question No. 1*** *is compulsory and choose* ***any four*** *questions from the remaining.*

1. Answer the following questions. [1x15=15]
   (a) What are the major types the control organization?
   (b) Give any two symbols used in register transfer language.
   (c) How many fields are there in an instruction?
   (d) What is the advantage of two wire handshaking method over strobe control method for asynchronous data transfer?
   (e) Is there any advantage of cycle stealing?
   (f) What is bootstrap loader?
   (g) What is a miss?
   (h) What is the function of logic microoperation?
   (i) Give any one feature of vectored interrupt.
   (j) What is an attached array processor?
   (k) What is parallel processing?
   (l) What is program counter (PR)?
   (m) What is priority interrupt?
   (n) What is a segment?
   (o) What is mapping process?

2. (a) Describe memory transfer. [4]
   (b) Draw a 4-bit arithmetic circuit and give the function table of the arithmetic circuit also. [4+3]

(c) Draw the flowchart for interrupt cycle. [4]

3. (a) Describe the common bus system (of basic computer) that connects the general purpose registers and memory for transferring of information. [10]

(b) Describe about the advantages and disadvantages of microprogrammed control organization and hardwired control organization? [5]

4. (a) What is Input-Output Interface? Why are they included in digital computers? [5]

(b) Describe about asynchronous data transfer using two-wired handshaking method in either source-initiated or destination-initiated transfer. [5]

(c) Explain about isolated I/O and memory mapped I/O. [5]

5. (a) Describe parallel priority interrupt. [5]

(b) Draw a block diagram for DMA transfer in a computer system. [5]

(c) Describe Programmed I/O. [5]

6. (a) Explain Indirect Addressing Mode, One-Address Instruction Format and Zero-Address Instruction Format. [3x3=9]

(b) Write an assembly language program to find the sum of a series of numbers. [6]

7. Write short notes on any five: [3x5=15]

(a) Auxiliary memory
(b) Paging
(c) Set-associative mapping
(d) RISC

(e) Segmentation
(f) Quad Core Processor
(g) Memory hierarchy
(h) Primary memory
(i) Indexed Addressing Mode